LECTURE NOTES ON ROTATION THEORY

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1. Introduction

Rotation theory is a field of Dynamical Systems that studies how the orbits of a system turn around certain topological or dynamical objects, e.g. "holes" or invariant sets of the phase space.

We can find its origin in the work of Poincaré [Poi80] where he studied the dynamics of circle homeomorphisms.

However, there is a very simple and classical example that can be considered in the *pre-history* of rotation theory: the study of the dynamics of vector fields on the circle.

In order to describe this simple example, let $\mathbb{T}$ denote the circle which is defined as the quotient space $\mathbb{R}/\mathbb{Z}$, and $\pi\colon \mathbb{R} \to \mathbb{T}$ be the natural quotient projection. Let $X\colon \mathbb{T} \to \mathbb{R}$ be a C^r-vector field on $\mathbb{T}$, with $r \geqslant 0$ and consider the flow $\Phi_X\colon \mathbb{R} \times \mathbb{T} \to \mathbb{T}$ generated by X, i.e. the function Φ_X is C^r and satisfies

$$\Phi_X^0(p) = p, \quad \text{and} \quad \frac{\partial}{\partial t}\Phi_X^t(p,t)\big|_{t=0} = X(p), \quad \forall p \in \mathbb{T}.$$

If X has singularities, i.e. $\mathsf{Sing}(X) := X^{-1}(0) \neq \varnothing$, then one can easily show that for every $p \in \mathbb{T}$ there is $p_-, p_+ \in \mathsf{Sing}(X)$ such that $\Phi_X^{\pm t}(p) \to p_\pm$, as $t \to +\infty$.

On the other hand, if we suppose that X is *singularity free,* i.e. $\mathsf{Sing}(X) = \varnothing$, then by uniqueness of solutions of ODEs we know that the flow Φ_X is periodic, i.e. there exists $T > 0$ such that $\Phi_X^{t+T}(p) = \Phi_X^t(p)$, for every $p \in \mathbb{T}$ and any $t \in \mathbb{R}$.

On the other hand, one can define the real number

$$\rho_X := \left(\int_{\mathbb{T}} \frac{dx}{X(x)}\right)^{-1}, \tag{1}$$

where dx denotes the Haar measure (a.k.a. the Lebesgue measure) on $\mathbb{T}$, and then the map $\tilde{h}\colon \mathbb{R} \to \mathbb{R}$ given by

$$\tilde{h}(\tilde{x}) := \rho_X \int_0^{\tilde{x}} \frac{dt}{X\big(\pi(t)\big)}, \quad \forall \tilde{x} \in \mathbb{R}.$$

Observe that $\tilde{h}$ is a strictly monotone C^{r+1} bijective map and satisfies $\tilde{h}(\tilde{x}+1) = \tilde{h}(\tilde{x}) + 1$, for all $\tilde{x} \in \mathbb{R}$. So, it is the lift of a C^{r+1} circle diffeomorphism $h\colon \mathbb{T} \circlearrowleft$, i.e. it holds $\pi \circ \tilde{h} = h \circ \pi$. So, one can easily verify that the *push-forward vector field* $h_\star X\colon \mathbb{T} \to \mathbb{R}$ given by

$$h_\star X(p) := Dh_{h^{-1}(p)}\Big(X\big(h^{-1}(p)\big)\Big), \quad \forall p \in \mathbb{T},$$

satisfies $h_\star X \equiv \rho_X$.

Exercise 1. *If X is as above and $\tilde{X}\colon \mathbb{R} \to \mathbb{R}$ is the lift of X, i.e. it holds $\tilde{X} \circ \pi = X$, then show that holds*

$$\frac{\Phi_{\tilde{X}}^t(\tilde{x}) - \tilde{x}}{t} \to \rho_X, \quad \textit{as } t \to \infty,$$

where the convergence is uniform in $\tilde{x} \in \mathbb{R}$.

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Exercise 2. *Let $X, Y\colon \mathbb{T} \to \mathbb{R}$ be two C^r singularity free vector fields. Show that they are C^{r+1}-conjugate (i.e. there is a C^{r+1}-diffeomorphism $h\colon \mathbb{T} \circlearrowleft$ satisfying $h_\star X = Y$), if and only if $\rho_X = \pm\rho_Y$.*

2. Circle homeomorphisms

The work of Poincaré [Poi80] extends the results about vector fields on the circle we exposed in §1 to circle homeomorphisms. Here we continue considering the circle $\mathbb{T} := \mathbb{R}/\mathbb{Z}$ and writing $\pi\colon \mathbb{R} \ni \tilde{x} \mapsto \tilde{x} + \mathbb{Z} \in \mathbb{R}/\mathbb{Z}$ for the natural quotient projection. We shall always assume $\mathbb{T}$ endowed with the distance function $d_{\mathbb{T}}\colon \mathbb{T} \times \mathbb{T} \to [0,1]$ given by

$$d_{\mathbb{T}}(x,y) := \inf\{|\tilde{x} - \tilde{y}| : \tilde{x} \in \pi^{-1}(x),\ \tilde{y} \in \pi^{-1}(y)\}, \quad \forall x, y \in \mathbb{T}.$$

Given any continuous map $f\colon \mathbb{T} \circlearrowleft$, **a lift** of f is a continuous map $\tilde{f}\colon \mathbb{R} \circlearrowleft$ such that $\pi \circ \tilde{f} = f \circ \pi$. The **degree of** f is defined as the natural number

$$\deg(f) := \tilde{f}(\tilde{x} + 1) - \tilde{f}(\tilde{x}) \in \mathbb{Z},$$

where one can easily show that this number is independent of $\tilde{x} \in \mathbb{R}$ and the chosen lift $\tilde{f}$.

If $f\colon \mathbb{T} \circlearrowleft$ is a homeomorphism, then one can show that $\deg(f) \in \{-1, 1\}$. We write $\mathrm{Homeo}(\mathbb{T})$ for the group of circle homeomorphisms, and we also define

$$\mathrm{Homeo}_+(\mathbb{T}) := \{f \in \mathrm{Homeo}(\mathbb{T}) : \deg(f) = 1\},$$
$$\mathrm{Homeo}_-(\mathbb{T}) := \{f \in \mathrm{Homeo}(\mathbb{T}) : \deg(f) = -1\}.$$

The elements of $\mathrm{Homeo}_+(\mathbb{T})$ are said to be **orientation preserving** homeomorphisms, and the elements of $\mathrm{Homeo}_-(\mathbb{T})$ are called **orientation reversing** homeomorphisms.

Exercise 3. *Let $f \in \mathrm{Homeo}(\mathbb{T})$ be a homeomorphism and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ be a lift of f. Show that f is orientation preserving if and only if $\tilde{f}$ is a strictly increasing homeomorphism. On the other hand, f is orientation reversing if and only if $\tilde{f}$ is a strictly decreasing homeomorphism.*

Exercise 4. *Give an example of an orientation preserving diffeomorphism $f\colon \mathbb{T} \circlearrowleft$ such that there is no vector field $X\colon \mathbb{T} \to \mathbb{R}$ such that $f = \Phi_X^t$, for any $t \in \mathbb{R}$.*

The dynamics of orientation reversing homeomorphisms can be easily described. In fact, if $f \in \mathrm{Homeo}_-(\mathbb{T})$ and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ is a lift of f, then $\tilde{f}_k := \tilde{f} + k$ is another lift, for every $k \in \mathbb{Z}$. One can easily show that $\sharp\,\mathrm{Fix}(\tilde{f}_k) = 1$, for every $k \in \mathbb{Z}$ and $\pi(\mathrm{Fix}(\tilde{f}_k)) = \pi\big(\mathrm{Fix}(\tilde{f}_\ell)\big)$, if and only if $k - l$ is an even integer. In particular, this implies that $\mathrm{Fix}(f)$ has exactly two fixed points and $\mathrm{Per}(f) = \mathrm{Fix}(f^2)$, noticing that $f^2 \in \mathrm{Homeo}_+(\mathbb{T})$. Finally, one can show that the α- and ω-limit of every point of $\mathbb{T}$ are two different periodic orbits.

So, from now on we shall concentrate in the study of orientation preserving homeomorphisms. Let us start considering two fundamental families of orientation-preserving circle diffeomorphisms:

Example 2.1 (Rigid rotations)**.** The circle $\mathbb{T} = \mathbb{R}/Z$ has a natural (Lie) group structure and hence, for every $\alpha \in \mathbb{T}$ one can consider the **rotation** (or translation) $R_\alpha \in \mathrm{Diff}^\infty_+(\mathbb{T})$ given by

$$R_\alpha(x) := x + \alpha, \quad \forall x \in \mathbb{T}.$$

The point α is said to be **rational** when $\alpha \in \mathbb{Q}/Z$, or equivalently, when $\pi^{-1}(\alpha) \subset \mathbb{Q}$. Otherwise, α is said to be **irrational** and this happends when $\pi^{-1}(\alpha) \subset \mathbb{R}\backslash\mathbb{Q}$.

One can show that α is rational if and only if there is $q \in \mathbb{N}$ such that $R_\alpha^q = id_{\mathbb{T}}$. And on the other hand, α is irrational if and only if R_α is **minimal**, i.e. $\varnothing$ and $\mathbb{T}$ are the only compact R_α-invariant sets.

Example 2.2 (Möbious transformations)**.** Let $\mathbb{RP}^1 := \mathbb{R}^2\backslash\{0\}/\sim$ be the **real projective line,** the quotient space induced by the equivalence relation on $\mathbb{R}^2\backslash\{0\}$ given by $v \sim w$ iff there is $\lambda \in \mathbb{R}\backslash\{0\}$ such that $\lambda v = w$. Observe that the map $D\colon \mathbb{T} \to \mathbb{RP}^1$ given by

$$D(t + \mathbb{Z}) := [\cos(2t\pi), \sin(2t\pi)], \quad \forall t \in \mathbb{R},$$

where $[v] \in \mathbb{RP}^1$ denotes the equivalence class containing $v \in \mathbb{R}^2\backslash\{0\}$, is indeed smooth diffeomorphism.

On the other hand, given any $A \in \mathrm{SL}_2(\mathbb{R})$ clearly induces a **projective map,** $f_A\colon \mathbb{RP}^1 \circlearrowleft$ given by $f_A[v] := [Av]$, for every $v \in \mathbb{R}^2\backslash\{0\}$. For us, a **Möbious transformation** will be any diffeomorphism $D^{-1} \circ f_A \circ D \in \mathrm{Diff}^\infty_+(\mathbb{T})$, with $A \in \mathrm{SL}_2(\mathbb{R})$.

Exercise 5 (Dynamics of Möbious transformations)**.** *Let $A \in \mathrm{SL}_2(\mathbb{R})$ and $h_A := D^{-1} \circ f_A \circ D$ be the corresponding Möbious transformation. Prove the following claims:*

(1) $h_A = id_{\mathbb{T}}$ iff $A \in \{I, -I\}$;
(2) h_A is conjugate to a rigid rotation iff $|\mathrm{tr}\, A| < 2$;
(3) h_A has exactly 2 fixed points and both of them are hyperbolic iff $|\mathrm{tr}\, A| > 2$;
(4) h_A has exactly one fixed point, which is parabolic, iff $|\mathrm{tr}\, A| = 2$ and $A \notin \{I, -I\}$.

Given any $f \in \mathrm{Homeo}_+(\mathbb{T})$ and a lift $\tilde{f}\colon \mathbb{R} \circlearrowleft$, one defines the **displacement function** or **displacement cocycle**

$$\Delta_{\tilde{f}} := \tilde{f} - id_{\mathbb{R}},$$

and it can be easily showed that the continuous function $\Delta_{\tilde{f}}$ is $\mathbb{Z}$-periodic, and hence, it can be considered as an element of $C^0(\mathbb{T}, \mathbb{R})$. Moreover, we say that this function is *cocycle* because it holds

$$\Delta_{\tilde{f}^n} = \tilde{f}^n - id_{\mathbb{R}} = \sum_{j=0}^{n-1} \Delta_{\tilde{f}} \circ f^j. \tag{2}$$

We shall write $\widetilde{\mathrm{Homeo}}_+(\mathbb{T})$ to denote the group of homeomorphisms of $\mathbb{R}$ that are lifts of orientation preserving circle homeomorphisms. One can verify that

$$\widetilde{\mathrm{Homeo}}_+(\mathbb{T}) := \left\{\tilde{f} \in \mathrm{Homeo}_+(\mathbb{R}) : \tilde{f} - id_{\mathbb{R}} \text{ is } \mathbb{Z}\text{-periodic}\right\}.$$

The next result allows us to introduce the main dynamical invariant in the study of circle homeomorphisms:

Theorem 2.3 (Rotation number)**.** *If $f \in \mathrm{Homeo}_+(\mathbb{T})$ and $\tilde{f} \in \widetilde{\mathrm{Homeo}}_+(\mathbb{T})$ is a lift of f, then there exists $\rho(\tilde{f}) \in \mathbb{R}$ such that*

$$\frac{\tilde{f}^n(\tilde{x}) - \tilde{x}}{n} \to \rho(\tilde{f}) := \int_{\mathbb{T}} \Delta_{\tilde{f}}\, \mathrm{d}\mu, \quad \text{as } n \to +\infty,$$

for every $x \in \mathbb{T}$, every $\tilde{x} \in \pi^{-1}(x)$, and for every $\mu \in \mathfrak{M}(f)$, where $\mathfrak{M}(f)$ denotes the space of f-invariant Borel probability measures. Thus, the convergence is uniform in $x \in \mathbb{T}$.

Before proving Theorem 2.3, we need the following simple lemma that will play a crucial role in the proof of subsequent results:

Lemma 2.4 (Oscillation of $\Delta_{\tilde{f}}$)**.** *If $f \in \mathrm{Homeo}_+(\mathbb{T})$ and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ is a lift of f, then it holds*

$$\operatorname{osc} \Delta_{\tilde{f}} := \max_{x\in\mathbb{T}} \Delta_{\tilde{f}}(x) - \min_{x\in\mathbb{T}} \Delta_{\tilde{f}}(x) < 1.$$

Proof of Lemma 2.4. Since $\Delta_{\tilde{f}}$ is $\mathbb{Z}$-periodic when considered as a real function defined on $\mathbb{R}$, then there exists numbers $\tilde{x}_m, \tilde{x}_M \in \mathbb{R}$ such that $\tilde{x}_m < \tilde{x}_M < \tilde{x}_m + 1$ and $\Delta_{\tilde{f}}(\tilde{x}_m) = \min \Delta_{\tilde{f}}$ and $\Delta_{\tilde{f}}(\tilde{x}_M) = \max \Delta_{\tilde{f}}$. Reasoning by contradiction, let us suppose that $\operatorname{osc} \Delta_{\tilde{f}} \geqslant 1$. Then it holds

$$\tilde{f}(\tilde{x}_M) - \tilde{f}(\tilde{x}_m) = \tilde{x}_M - \tilde{x}_m + \Delta_{\tilde{f}}(\tilde{x}_M) - \Delta_{\tilde{f}}(\tilde{x}_m) > 1.$$

Thus, there exists $\tilde{y} \in \mathbb{R}$ such that $\tilde{f}(\tilde{x}_m) < \tilde{y} < \tilde{y} + 1 < \tilde{f}(\tilde{x}_M)$. By continuity of $\tilde{f}$, there exists two points $\tilde{x}_0, \tilde{x}_1 \in \mathbb{R}$, such that $\tilde{x}_m < \tilde{x}_0 < \tilde{x}_1 < \tilde{x}_M$, and $\tilde{f}(\tilde{x}_0) = \tilde{y}$ and $\tilde{f}(\tilde{x}_1) = \tilde{y} + 1$. Then, we have

$$f(x_i) = \pi\big(\tilde{f}(\tilde{x}_i)\big) = \pi(\tilde{y} + i) = \pi(\tilde{y}),$$

for $i = 0, 1$, and where $x_i := \pi(\tilde{x}_i)$, but since $|\tilde{x}_1 - \tilde{x}_0| < \tilde{x}_M - \tilde{x}_m < 1$, it holds $x_0 \neq x_1$, contradicting the injectivity of f. □

Proof of Theorem 2.6. Let ν be any ergodic f-invariant probability measure. If we define

$$\rho(\tilde{f}) := \int_{\mathbb{T}} \Delta_{\tilde{f}} \, \mathrm{d}\nu,$$

by Birkhoff ergodic theorem, we know there exists $x_0 \in \mathbb{T}$ such that

$$\frac{\tilde{f}^n(\tilde{x}_0) - \tilde{x}_0}{n} = \frac{1}{n} \sum_{j=0} \Delta_{\tilde{f}}\big(f^j(x_0)\big) \to \rho(\tilde{f}), \quad \text{as } n \to +\infty,$$

and for every $\tilde{x}_0 \in \pi^{-1}(x_0)$.

On the other hand, as a consequence of Lemma 2.4, we know that

$$\operatorname{osc} \Delta_{\tilde{f}^n} \leqslant 1, \quad \forall n \in \mathbb{Z}.$$

So, it holds

$$\left| \frac{\tilde{f}^n(\tilde{y}) - \tilde{y}}{n} - \frac{\tilde{f}^n(\tilde{x}_0) - \tilde{x}_0}{n} \right| \leqslant \frac{\operatorname{osc} \Delta_{\tilde{f}^n}}{n} \leqslant \frac{1}{n} \to 0,$$

as $n \to +\infty$, and the independence of the point and the uniformity of the convergence are proven.

Then, by Birkhoff ergodic theorem we get

$$\rho(\tilde{f}) = \int_{\mathbb{T}} \Delta_{\tilde{f}} \, \mathrm{d}\mu, \quad \forall \mu \in \mathfrak{M}(f).$$

□

As consequence of Lemma 2.4 and Theorem 2.3, we get the following

Corollary 2.5 (Bounded rotational deviations)**.** *If f, $\tilde{f}$ and $\rho(\tilde{f})$ are as in Theorem 2.3, then for every $n \in \mathbb{Z}$ there is $x_n \in \mathbb{T}$ such that*

$$\Delta_{\tilde{f}^n}(x_n) = n\rho(\tilde{f}),$$

and

$$\left|\tilde{f}^n(z) - z - n\rho(\tilde{f})\right| \leqslant 1, \quad \forall z \in \mathbb{R}, \tag{3}$$

and every $n \in \mathbb{Z}$.

Proof. Let μ be an arbitrary f-invariant measure. Then, by the cocycle condition (2) we get

$$\int_{\mathbb{T}} \Delta_{\tilde{f}^n} \, \mathrm{d}\mu = n \int_{\mathbb{T}} \Delta_{\tilde{f}} \, \mathrm{d}\mu, \quad \forall n \in \mathbb{Z},$$

and hence, there exists $x_n \in \mathbb{T}$ with

$$\Delta_{\tilde{f}^n}(x_n) = n \int_{\mathbb{T}} \Delta_{\tilde{f}} \, \mathrm{d}\mu = n\rho(\tilde{f}), \quad \forall n \in \mathbb{Z}.$$

Finally, invoking Lemma 2.4, we get

$$\left|\tilde{f}^n(\tilde{z}) - \tilde{z} - n\rho(\tilde{f})\right| = \left|\Delta_{\tilde{f}^n}(z) - \Delta_{\tilde{f}^n}(x_n)\right| \leqslant 1, \quad \forall n \in \mathbb{Z}, \ \forall \tilde{z} \in \mathbb{R},$$

and where $z = \pi(\tilde{z})$. □

Exercise 6 (Continuity of ρ)**.** *Prove that* $\rho\colon \widetilde{\mathrm{Homeo}}_+(\mathbb{T}) \to \mathbb{R}$ *is a continuous map. Hint: use estimate* (3)

Then we have the following result relating the rationality of $\rho(\tilde{f})$ and the existence of periodic points for f:

Theorem 2.6 (Existence of periodic points)**.** *Let* $f \in \mathrm{Homeo}_+(\mathbb{T})$ *and* $\tilde{f}\colon \mathbb{R} \circlearrowleft$ *be a lift of* f*. Then, if* $z \in \mathrm{Per}(f)$ *is a periodic point of* ***prime period*** $q \in \mathbb{N}$ *(i.e.* $f^q(z) = z$ *and* $f^i(z) \neq z$ *for* $i = 1, 2, \ldots, q-1$*), then every periodic point has prime period* q *and there is an integer number* $p \in \mathbb{Z}$*, with* $\gcd(p, q) = 1$ *and such that*

$$\tilde{f}^q(\tilde{x}) = \tilde{x} + p, \quad \forall x \in \mathrm{Per}(f), \ \forall \tilde{x} \in \pi^{-1}(x).$$

In such a case, it clearly holds $\rho(\tilde{f}) = p/q$*.*

Reciprocally, if f *is* ***periodic point free,*** *i.e.* $\mathrm{Per}(f) = \varnothing$*, then* $\rho(\tilde{f}) \in \mathbb{R}\backslash\mathbb{Q}$*.*

Proof. Let us start assuming that $\rho(\tilde{f}) = p/q \in \mathbb{Q}$, with $p, q \in \mathbb{Z}$ are coprime. Then, by Corollary 2.5, there is a point $x_q \in \mathbb{T}$ such that

$$\Delta_{\tilde{f}^q}(x_q) = q\rho(\tilde{f}) = p \in \mathbb{Z}.$$

So, $\tilde{f}^q(\tilde{x}_q) = \tilde{x}_q + p$, for any $\tilde{x}_q \in \pi^{-1}(x_q)$ and thus, $x_q \in \mathrm{Per}(f)$.

On the other hand, let us suppose $\mathrm{Per}(f) \neq \varnothing$. Then, given any $z \in \mathrm{Per}(f)$ with prime period q, there exists an integer number $p \in \mathbb{Z}$ such that

$$\tilde{f}^q(\tilde{z}) = \tilde{z} + p, \quad \forall \tilde{z} \in \pi^{-1}(z).$$

This clearly implies that $\rho(\tilde{f}) = p/q$, and by the dynamics of orientation preserving interval maps, one can easily verifies that any other periodic point must have prime period q. Finally, if p and q were not coprime, writing $\rho(\tilde{f}) = p'/q'$ with $p' \in \mathbb{Z}$ and $q' \in \mathbb{N}$ coprime numbers, by the previous argument one could find a periodic point of prime period q', with $0 < q' < q$, contradicting the fact that every periodic point has prime period q. □

On the other hand, we have a rather good description of the dynamics of circle homeomorphisms with irrational rotation number. Let us start with a simple

Proposition 2.7 (Uniqueness of minimal set)**.** *Let* $f \in \mathrm{Homeo}_+(\mathbb{T})$ *be a periodic point free homeomorphism. Then, the non-wandering set* $\Omega(f) \subset \mathbb{T}$ *is the only minimal set for* f *and every connected component of* $\mathbb{T}\backslash\Omega(f)$ *is a wandering set for* f*.*

Proof. By Zorn's lemma, there exists a minimal set $K \subset \mathbb{T}$ for f. If $K = \mathbb{T}$, the it is clearly the only minimal set and $K = \Omega(f)$. On the other hand, if $K \neq \mathbb{T}$, let I be a connected component of the open set $\mathbb{T}\backslash K$. Since f is periodic point free, if there is $n \geqslant 1$ such that $f^n(I) \cap I \neq \varnothing$, then it holds indeed $f^n(I) = I$ and the extreme points of I are periodic for f, contradicting the fact that f is periodic point free. So, every connected component of $\mathbb{T}\backslash K$ is a wandering domain for f and hence, $\Omega(f) \subset K$. By the minimality of K, we conclude that $K = \Omega(f)$ and thus, $\Omega(f)$ is the only minimal set. □

Theorem 2.8 (Irrational rigid rotations as topological factors)**.** *Let $f \in \mathrm{Homeo}_+(\mathbb{T})$ be a periodic point free homeomorphism, $\tilde{f}\colon \mathbb{R} \circlearrowleft$ be a lift of f and $\rho := \rho(\tilde{f}) \in \mathbb{R}\backslash\mathbb{Q}$ be its rotation number. Then, the following dichotomy holds:*

(a) either $\Omega(f) = \mathbb{T}$ and f is topologically conjugate to the rigid rotation $R_\rho\colon \mathbb{T}$;
(b) or $\Omega(f)$ is a Cantor set, and there is degree-1 continuous map $h\colon \mathbb{T} \circlearrowleft$ satisfying $h \circ f = R_\rho \circ h$. Moreover, $h^{-1}(x) \subset \mathbb{T}$ is either the closure of a connected component of $\mathbb{T}\backslash\Omega(f)$, or a unique point of $\Omega(f)$ which is not the boundary point of a gap of $\Omega(f)$ (i.e. a connected component of $\mathbb{T}\backslash\Omega(f)$).

Before proving Theorem 2.8, we need a preliminary result:

Theorem 2.9 (Gottschalk-Hedlund theorem)**.** *Let X be a compact metric space, $f\colon X \circlearrowleft$ be a homeomorphism and $\phi\colon X \to \mathbb{R}$ be a continuous function. Then, there exists $u \in C^0(X, \mathbb{R})$ satisfying*

$$u \circ f - u = \phi,$$

if and only if there is a point $x_0 \in X$ such that

$$\sup_{n\in\mathbb{N}} \left| \sum_{j=0}^{n-1} \phi\big(f(x_0)\big) \right| < \infty. \tag{4}$$

Proof of Theorem 2.9. Let us start assuming there is $u \in C^0(X, \mathbb{R})$ such that $u \circ f - u = \phi$. Then, given any $x \in X$, it holds

$$\left| \sum_{j=0}^{n-1} \phi\big(f^j(x)\big) \right| = \big|u\big(f^n(x)\big) - u(x)\big| \leqslant \|\phi\|, \quad \forall n \in \mathbb{N},$$

where the norm $\|\phi\| := \sup_{y\in X} |\phi(y)|$ is clearly finite.

Reciprocally, let us suppose there exists $x_0 \in X$ such that estimate (4) holds, and consider the homeomorphism $F\colon X \times \mathbb{R} \circlearrowleft$ given by

$$F(x,t) := \big(f(x), t - \phi(x)\big), \quad \forall (x,t) \in X \times \mathbb{R}.$$

By (4) we know that the F-orbit of the point $(x_0, 0)$, which is denoted by $\mathcal{O}_F(x_0, 0)$, is bounded. Then, there exists a minimal set $M \subset \overline{\mathcal{O}_F(x_0, 0)}$. By the minimality of f, we know that $\mathrm{pr}_1(M) = X$, where $\mathrm{pr}_1\colon X \times \mathbb{R} \to X$ denotes the projection on the first coordinate. On the other hand, we claim that the M is indeed the graph of a continuous function $u\colon X \to \mathbb{R}$. To prove that, reasoning by contradiction let us suppose there are $x_1 \in X$, $t, t' \in \mathbb{R}$ with $t \neq t'$, such that $(x_1, t), (x_1, t') \in X \times \mathbb{R}$. Let us consider the homeomorphism $T\colon X \times \mathbb{R} \ni (x,s) \mapsto (x, s + t - t') \in X \times \mathbb{R}$ and observe that $F \circ T = T \circ F$. This implies that $T(M)$ is a minimal set for F, too. But since $(x,t) \in M \cap T(M)$, by minimality of M and $T(M)$ we conclude that $M = T(M)$. So, $M = T^n(M)$ for every $n \in \mathbb{Z}$ and since $t - t' \neq 0$, we conclude that M is unbounded, contradicting the fact that was compact.

Then, M is a compact set which is the graph of a function $u\colon X \to \mathbb{R}$. Since, X and M are compact, we conclude that u is continuous. Finally, since M is F-invariant, it clearly holds $F\big(x, u(x)\big) = \big(f(x), u\big(f(x)\big) - \phi(x)\big)$, for every $x \in X$. □

Proof of Theorem 2.8. Since $\Omega(f)$ is a minimal set for f and its lift $\tilde{f}$ satisfies estimate (3), we can apply Theorem 2.9 to guarantee the existence of a continuous function $u\colon \Omega(f) \to \mathbb{R}$ such that

$$u\big(f(x)\big) - u(x) = \rho - \Delta_{\tilde{f}}(x), \quad \forall x \in \Omega(f).$$

Now, consider the function $\tilde{h}\colon \pi^{-1}\big(\Omega(f)\big) \subset \mathbb{R} \to \mathbb{R}$ given by

$$\tilde{h}(\tilde{x}) := \tilde{x} + u(\tilde{x}), \quad \forall \tilde{x} \in \pi^{-1}\big(\Omega(f)\big).$$

One can easily check that

$$\tilde{h}\big(\tilde{f}(\tilde{x})\big) = \tilde{h}(\tilde{x}) + \rho, \quad \forall \tilde{x} \in \pi^{-1}\big(\Omega(f)\big),$$

and so, there is a continuous map $h\colon \Omega(f) \to \mathbb{T}$ satisfying $h \circ f(x) = R_\rho \circ h(x)$, for all $x \in \Omega(f)$.

If $\Omega(f) = \mathbb{T}$, then h is indeed a degree-1 continuous semi-conjugacy between f and R_ρ, and it just remains to prove its monotonicity. On the other hand, let us suppose $\Omega(f) \neq \mathbb{T}$, and let I be a connected component of $\mathbb{T}\backslash\Omega(f)$. Since I is a wandering domain for f, it clearly holds

$$\sum_{n\in\mathbb{Z}} \mathrm{Leb}\big(f^n(I)\big) \leqslant 1, \tag{5}$$

and thus, $\mathrm{Leb}\big(f^n(I)\big) \to 0$, as $n \to \pm\infty$. So, if write $a, b \in \overline{I}$ for the boundary points of I, we get

$$d_{\mathbb{T}}\big(h(a), h(b)\big) = d_{\mathbb{T}}\Big(R_\rho^n\big(h(a)\big), R_\rho^n\big(h(b)\big)\Big) = d_{\mathbb{T}}\Big(h\big(f^n(a)\big), h\big(f^n(b)\big)\Big) \to 0,$$

as $n \to \pm\infty$. So, $h(a) = h(b)$ and we can continuously extends h to the whole circle $\mathbb{T}$ just defining h as a constant function on each connected component of $\mathbb{T}\backslash\Omega(f)$. Again in this case we get a degree-1 continuous semi-conjugacy between f and R_ρ. In order to finish the proof, we need the following

Exercise 7 (Semi-conjugacies with irrational rotations)**.** *Let $f \in \mathrm{Homeo}_+(\mathbb{T})$ be a periodic point free homeomorphism, $\tilde{f}\colon \mathbb{R} \circlearrowleft$ any lift of f and $\rho := \rho(\tilde{f})$. Then, if $h\colon \mathbb{T} \circlearrowleft$ is a degree-1 continuous map satisfying $h \circ f = R_\rho \circ h$, then, the following properties hold:*

(1) h is injective on $\Omega(f)$;
(2) h is constant on every wandering open interval;
(3) h is monotone, i.e. every lift $\tilde{h}\colon \mathbb{R} \circlearrowleft$ is an increasing function.

Consequently, h is a homeomorphism if and only if $\Omega(f) = \mathbb{T}$.

□

Exercise 8 (Unique ergodicity)**.** *Show that every periodic point free circle homeomorphism is uniquely ergodic.*

Finally, we state the following

Theorem 2.10 (Monotonicity of the rotation number)**.** *Let $f \in \mathrm{Homeo}_+(\mathbb{T})$ be an arbitrary homeomorphism and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ be a lift of f. For each $a \in \mathbb{R}$ consider the homeomorphism $\tilde{f}_a(z) := \tilde{f}(z) + a$, for all $z \in \mathbb{R}$ and consider the real function $\rho_{\tilde{f}}\colon \mathbb{R} \to \mathbb{R}$ given by*

$$\rho_{\tilde{f}}(a) := \rho\big(\tilde{f}_a\big), \quad \forall a \in \mathbb{R}.$$

Then, $\rho_{\tilde{f}}$ is an increasing function satisfying $\rho_{\tilde{f}}(n) = \rho_{\tilde{f}}(0) + n$, for every $n \in \mathbb{Z}$ and such that $\rho_{\tilde{f}}$ is locally constant at the point $a_0 \in \mathbb{R}$ if and only if $\rho_{\tilde{f}}(a_0) \in \mathbb{Q}$ and $f_{a_0}^n \neq id_{\mathbb{T}}$, for every $n \in \mathbb{Z}\backslash\{0\}$, where $f_{a_0} \in \mathrm{Homeo}_+(\mathbb{T})$ is the circle homeomorphism that has $\tilde{f}_{a_0}$ as a lift.

2.1. **Circle diffeomorphisms.** The role of smoothness in the study of dynamical systems on the circle is very well understood. The firs result about circle diffeomorphisms is due to Denjoy:

Theorem 2.11 (Denjoy theorem)**.** *If $f\colon \mathbb{T} \circlearrowleft$ is an orientation preserving periodic point free $C^{1+\mathrm{vb}}$-diffeomorphism (i.e. f is a C^1 diffeomorphism and its derivative Df has bounded variation), then $\Omega(f) = \mathbb{T}$ and consequently, f is topologically conjugate to an irrational circle rotation.*

Recalling the concept of *bounded variation,* we have

Definition 1 (Bounded variation functions)**.** Given a compact interval $[a,b] \subset \mathbb{R}$, we denote its set of **finite partitions** $\mathcal{P}[a,b]$ which is given by

$$\mathcal{P}[a,b] := \left\{P = \{x_0, x_1, \dots, x_{n_P}\} : n_P \in \mathbb{N},\ x_0 = a < x_1 < x_2 < \dots < x_{n_P} = b\right\}.$$

A function $\phi\colon [a,b] \to \mathbb{R}$ is of **bounded variation** when it holds

$$V_a^b \phi := \sup_{P=\{x_i\}\in\mathcal{P}[a,b]} \sum_{i=0}^{n_P-1} |\phi(x_{i+1}) - \phi(x_i)| < \infty$$

Finally a C^0 function $\phi\colon \mathbb{T} \to \mathbb{R}$ is of **bounded variation** when $\tilde{\phi}|_{[0,1]}\colon [0,1] \to \mathbb{R}$ is of bounded variation, for some (and hence every) lift $\tilde{\phi}\colon \mathbb{R} \to \mathbb{R}$ (i.e. $\tilde{\phi}$ is continuous and $\phi \circ \pi = \tilde{\phi}$). In such a case, one writes $V_{\mathbb{T}}\phi := V_0^1 \tilde{\phi}$.

Exercise 9. *Show that any function $\phi\colon [a,b] \to \mathbb{R}$ of bounded variation is continuous. On the other hand, if ϕ is Lipschitz, then it is of bounded variation and it holds*

$$V_a^b \phi \leqslant \operatorname{Lip}\phi (b-a),$$

where $\operatorname{Lip}\phi := \sup_{x\neq y} \frac{|\phi(x)-\phi(y)|}{|x-y|}$*; and if ϕ is C^1, then it holds it holds*

$$V_a^b \phi = \int_a^b \left|f'(x)\right| \,\mathrm{d}x.$$

Finally, prove that for every $\alpha \in (0,1)$ there is an α-Hölder function $\phi\colon [a,b] \to \mathbb{R}$ which is not of bounded variation.

On the other hand, we have

Theorem 2.12 (Denjoy counter-example)**.** *For every $\alpha \in (0,1)$, every $\rho \in \mathbb{R}\backslash\mathbb{Q}$ and every neighborhood $\mathcal{U}$ of the rigid rotation R_ρ in the space $\mathrm{Diff}^1_+(\mathbb{T})$, there is a $C^{1+\alpha}$-diffeomorphism $f \in \mathcal{U}$ such that f is a topological extension of R_ρ, but it is not topologically conjugate to R_ρ. In other words, $\Omega(f) \neq \mathbb{T}$ and there is a lift $\tilde{f}\colon \mathbb{R} \circlearrowleft$ of f such that $\rho(\tilde{f}) = \rho$.*

Proof of Theorem 2.11. See for instance [Nav11, Theorem 3.1.1]. □

Proof of Theorem 2.12. See for instance [Nav11, Theorem 3.1.2]. □

So, as consequence of Theorem 2.11, if $f \in \mathrm{Diff}^2_+(\mathbb{T})$ is a periodic point free diffeomorphism and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ is a lift of f, then f is topologically conjugate to the rigid rotation $R_\rho\colon \mathbb{T} \circlearrowleft$, with $\rho := \rho(\tilde{f})$. Now, if $h_1, h_2 \in \mathrm{Homeo}_+(\mathbb{T})$ are two conjugacies between f and R_ρ, i.e. $h_i \circ f = R_\rho \circ h_i$, for $i = 1,2$, then it clearly holds

$$h_1 \circ h_2^{-1} \circ R_\rho = h_1 \circ f \circ h_2^{-1} = R_\rho \circ h_1 \circ h_2^{-1}.$$

In particular, this implies that $h_1 \circ h_2^{-1}$ is a homeomorphism that commutes with R_ρ.

Exercise 10. *Let ρ be an arbitrary irrational number and $h \in \mathrm{Homeo}(\mathbb{T})$ be a homeomorphism such that $h \circ R_\rho = R_\rho \circ h$. Show that there is a real number α such that $h = R_\alpha$.*

As a consequence of Exercise 10 we get that there is $\alpha \in \mathbb{R}$ such that $h_1 = R_\alpha \circ h_2$, and this implies that h_1 and h_2 have the same regularity. So, from now we shall say that a periodic point free diffeomorphism $f \in \mathrm{Diff}^r_+(\mathbb{T})$ is **C^s-linearizable** when any homeomorphism conjugating f to the corresponding rigid rotation is indeed a C^s-diffeomorphism, typically with $r \in [2, \infty]$ and $0 \leqslant s \leqslant r$.

The smoothness of the linearization of periodic point free circle diffeomorphisms has a strong relation with some arithmetical properties of the corresponding rotation number. This is the main motivation to intruduce the following

Definition 2 (Diophantine numbers)**.** A number $\alpha \in \mathbb{R}$ is said to be **Diophantine** when there exist real constants $C, \tau > 0$ such that

$$\left|\alpha - \frac{p}{q}\right| \geqslant \frac{C}{q^\tau}, \quad \forall p \in \mathbb{Z},\ \forall q \in \mathbb{N}.$$

Observe that every Diophantine number is irrational. On the other hand, every irrational number that is not Diophantine, is said to be **Liouville**.

Then we have the following remarkable result which is a consequence of the works of Arnold, Herman [Her79] and Yoccoz [Yoc84]:

Theorem 2.13 (Smooth linearization of diffeomorphisms of the circle)**.** *Let $f \in \mathrm{Diff}^\infty_+(\mathbb{T})$ and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ be a lift of f such that $\rho = \rho(\tilde{f}) \in \mathbb{R}\backslash\mathbb{Q}$ is a Diophantine number. Then f is C^∞-linearizable.*

On the other hand, for every $\alpha \in \mathbb{R}\backslash\mathbb{Q}$ Liouville, there exist $g \in \mathrm{Diff}^\infty_+(\mathbb{T})$ and a lift $\tilde{g}\colon \mathbb{R} \circlearrowleft$ of g such that $\rho(\tilde{g}) = \alpha$ and g is not C^1-linearizable.

We won't give a complete proof of Theorem 2.13 here, but we shall disccuss some important ingredients of the proof that play a role higher dimensions.

First of all we have the *linearized version* of Theorem 2.13:

Exercise 11 (Solving cohomological equations)**.** *Let $\alpha \in \mathbb{R}\backslash\mathbb{Q}$ be an arbitrary Diophantine number. Show that for every $\phi \in C^\infty(\mathbb{T}, \mathbb{R})$, there is $K \in \mathbb{R}$ and $u \in C^\infty(\mathbb{T}, \mathbb{R})$ such that*

$$u \circ R_\alpha - u = \phi - K.$$

Moreover, there exists a real constant $C > 0$ and an $s \in \mathbb{N}$, both numbers depending just on α, such that $\|u\|_\ell \leqslant C\,\|\phi\|_{\ell+s}$, for every $\ell \in \mathbb{N}_0$, and where

$$\|\phi\|_\ell := \max_{0 \leqslant k \leqslant \ell} \max_{x \in \mathbb{T}} \left|D^k \phi(x)\right|, \quad \forall \phi \in C^\infty(\mathbb{T}, \mathbb{R}).$$

On the other hand, given any Liouville number $\alpha \in \mathbb{R}\backslash\mathbb{Q}$, show that there exists $\phi \in C^\infty(\mathbb{T}, \mathbb{R})$ such that $u \circ R_\alpha - u \neq \phi - K$, for every $u \in C^\infty(\mathbb{T}, \mathbb{R})$ and evey $K \in \mathbb{R}$.

Now we can state the main local linearization result:

Theorem 2.14 (Local linearization theorem)**.** *Let $\alpha \in \mathbb{R}\backslash\mathbb{Q}$ be a Diophantine number. Then there exists a neighborhood $\mathcal{U}$ of the rotation R_α in $\mathrm{Diff}^\infty_+(\mathbb{T})$ such that for every $f \in \mathcal{U}$, there exist $\lambda \in \mathbb{R}$ and $h \in \mathrm{Diff}^\infty_+(\mathbb{T})$ such that*

$$f = R_\lambda \circ h \circ R_\alpha \circ h^{-1}.$$

References

[Her79] M. Herman, *Sur la conjugaison différentiable des difféomorphismes du cercle à des rotations*, Inst. Hautes Études Sci. Publ. Math. **49** (1979), 5–233.

[Nav11] A. Navas, *Groups of circle diffeomorphisms*, spanish ed., Chicago Lectures in Mathematics, University of Chicago Press, Chicago, IL, 2011. MR 2809110

[Poi80] H. Poincaré, *Mémoire sur les courbes définies par les équations différentielles I–VI, oeuvre I*, Gauthier-Villar: Paris (1880), 375–422.

[Yoc84] J. C. Yoccoz, *Conjugaison différentiable des difféomorphismes du cercle dont le nombre de rotation vérifie une condition diophantienne*, Annales scientifiques de l'É.N.S. **17** (1984), 333–359.

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